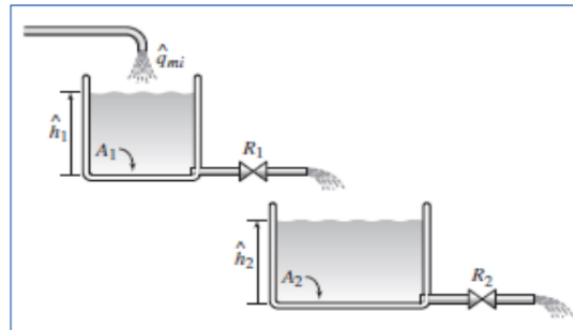


### Question (1): [10 Marks]

a) For the following liquid system:

- Derive the governing differential equations. (2 marks)
- Find the transfer function between  $q_i$  and  $h_2$  of the two tanks. (2 marks)
- Represent the transfer functions of the two tanks in a block diagram. (1 mark)



$$i) \quad \hat{q}_{mi} - \frac{h_1}{R_1} = A_1 \dot{h}_1 \xrightarrow{\times R_1} R_1 \hat{q}_{mi}(s) - H_1(s) = s A_1 R_1 H_1(s)$$

$$\frac{h_1}{R_1} - \frac{h_2}{R_2} = A_2 \dot{h}_2 \xrightarrow{\times R_1 R_2} R_2 H_1(s) - R_1 H_2(s) = A_2 R_1 R_2 s H_2(s)$$

$$ii) \quad R_1 \hat{q}_{mi}(s) = (s A_1 R_1 + 1) H_1(s) \rightarrow \frac{H_1(s)}{\hat{q}_{mi}(s)} = \frac{R_1}{s A_1 R_1 + 1}$$

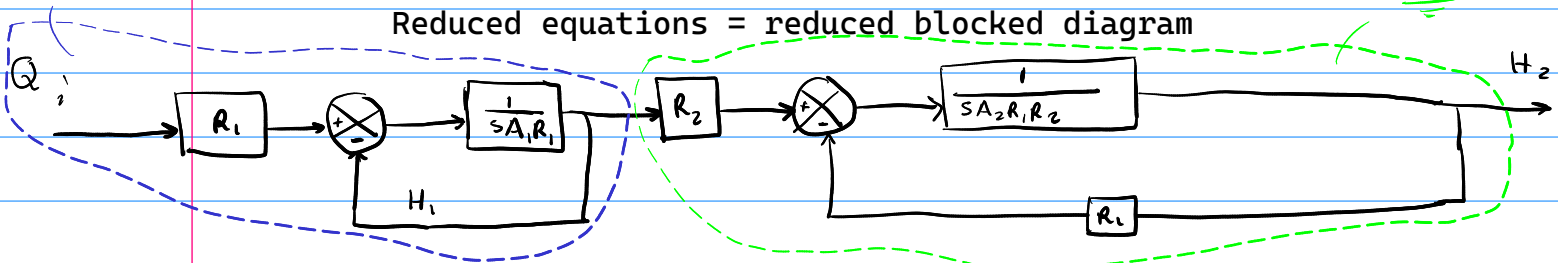
$$R_2 H_1(s) = (s A_2 R_1 R_2 + R_1) H_2(s) \rightarrow \frac{H_2(s)}{H_1(s)} = \frac{R_2}{s A_2 R_1 R_2 + R_1}$$

$$\frac{H_2(s)}{\hat{q}_{mi}(s)} = \frac{H_1(s)}{\hat{q}_{mi}(s)} \times \frac{H_2(s)}{H_1(s)} = \frac{R_1 R_2}{(s A_1 R_1 + 1)(s A_2 R_1 R_2 + R_1)}$$

TANK 1

Tank 2

Reduced equations = reduced blocked diagram



$$iii) \quad \frac{H_1(s)}{\hat{q}_{mi}(s)} = R_1 \cdot \frac{1}{s A_1 R_1 + 1} \rightarrow \frac{H_1(s)}{\hat{q}_{mi}(s)} = \frac{R_1}{s A_1 R_1 + 1}$$

$$\frac{H_1(s)}{z(s)} = R_1 \rightarrow h_1(t) = R_1 z$$

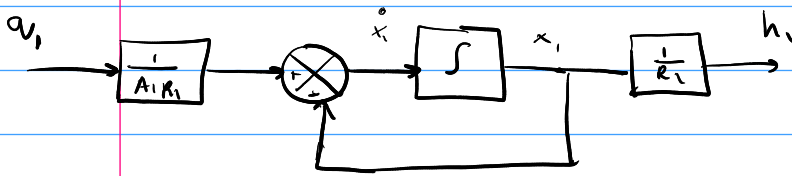
$$\frac{z(s)}{\hat{q}_{mi}(s)} = \frac{1}{s A_1 R_1 + 1} \rightarrow \dot{q}_i = A_1 R_1 \dot{z} + z$$

$$x_1 = z \rightarrow \dot{x}_1 = \dot{z}$$

$$h_1 = R_1 x_1 \quad \dot{x}_1 = \frac{1}{A_1 R_1} \dot{q}_i - \frac{1}{A_1 R_1} x_1$$

$$x_1 = z \rightarrow x_1^0 = 0$$

$$h_1 = R_1 x_1 \quad x_1^0 = \frac{1}{A_1 R_1} a_1 - \frac{1}{A_1 R_1} x_1$$



same for  $\frac{H_2(s)}{U_1(s)}$